# JalVaani

*Noise-Resilient Voice Intelligence for Coastal Communities*

**Every voice deserves to be heard. Every story deserves to be preserved.**

## The problem we're addressing

In coastal communities, spoken communication carries everyday information, local experience and stories passed between generations. But wind, boat engines and other background sounds can make recordings difficult to understand. Multilingual speech and important names, places or details can make transcripts harder to review. As a result, useful information and community knowledge can be difficult to document and preserve.

## Our approach

JalVaani is a voice-intelligence platform built around a simple idea: converting speech into text is only one part of making a recording useful. It uses Gnani AI speech-to-text capabilities and brings together tools to review transcripts, assess transcription quality, compare selected details with a reference, and save records for later access.

## Key features

- **Speech-to-Text —** Converts audio recordings into written transcripts using Gnani AI.
- **ShorProof —** Evaluates transcription quality using Word Error Rate (WER), Character Error Rate (CER) and processing latency; it supports comparisons between quiet and noisy recordings when reference transcripts are available.
- **ContextEcho —** Highlights phrases that may need attention so a person can review them rather than relying on an uncertain transcript.
- **CriticalDetails —** Helps compare important transcript information with a supplied reference.
- **YaadVault —** Allows transcript records to be saved and retrieved, supporting the documentation of stories and information.

## Who it is for and why it matters

JalVaani is designed with coastal communities and people working with multilingual spoken information in mind. Our aim is to make recordings easier to review and organise, while helping preserve stories and local knowledge in a more accessible form. Human review remains important, especially when a recording contains critical information.

## How we built it

Our two-member team worked across backend and frontend development. The backend uses Python and FastAPI, with Gnani AI speech-to-text integration and transcript-processing workflows. The frontend uses React and TypeScript, with browser-based audio recording, transcript editing, connectivity indicators and offline-related task handling. The project also includes transcript-quality evaluation and record-management features.

## Explore JalVaani

**Live project:** https://jalvaani-frontend.onrender.com/
**Source code:** https://github.com/Plansha/JalVaani